Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 8 · SLO 1 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 8 : SLO â€“ 1 | Queue â€” Introduction

I. Fill in the Blanks

• A queue follows the FIFO (First-In-First-Out) principle.
• In a queue, insertion is performed at the REAR end.
• In a queue, deletion is performed at the FRONT end.
• A SIMPLE (linear) queue is a linear data structure with a fixed size.
• A CIRCULAR queue overcomes the limitation of fixed size by wrapping around.
• A DEQUE (double-ended queue) allows insertion and deletion from both ends.
• PRIORITY queue gives priority to elements based on certain criteria.
• Queues are used in CPU (process/job) scheduling in operating systems.
• In a queue, if the rear is just behind the front in a circular manner, the queue is said to be FULL.
• An ARRAY (or linked list) is used to implement queues in programming languages like C.
• The queue operation that checks whether it is full is called ISFULL.
• The queue operation that checks whether it is empty is called ISEMPTY.
• In breadth-first search (BFS) of a graph, A QUEUE is used to keep track of the next node to visit.
• In printers, tasks are managed using a QUEUE to print in the order they arrive.
• A queue where new elements are added based on priority rather than position is called a PRIORITY QUEUE.